

STATS 506 Final Project Report

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Github Repo: <https://github.com/pdlekhaa/506-final-project>

1 Intro

Accountability is a necessary compliment to authority. Record keeping ensures rules are being followed and the law is justly applied to all citizens. I am investigating information pertaining to San Jose, which has public records of arrests, charges, and incidents.

There are three main datasets I am using, coming from the same Kaggle resource [1]. The main dataset is *arrests_all.csv* which contains information about arrests, as well as personal identifiers (age, sex, race). An arrest may or may not have (a) corresponding charge(s) from the *charges_all.csv*. Finally, an arrest may or may not have a corresponding incident from the *incidents_all.csv* dataset, which tracks which is the reported reason a police officer comes to the scene.

My research question is: What do we make of arrests that do not have corresponding charges? What groups of people are likely to fall into this category?

2 Methods

First I had to identify which rows are actually arrests that do not have a corresponding charges. Most could be matched though the same *PIN* and *GO.NO* (Personal Identification Number and Incident Number). However, this actually left many arrests unaccounted for, mainly due to blank *GO.NO* fields. Of these, rows were matched with charges through an updated *ARREST.DATE* and *PIN* if possible. Anything left were considered arrests without charges.

For comparison, arrests with corresponding charges were cleaned and combined; any incorrectly matched rows were removed (ie mismatched dates and an empty *GO.NO*), along with duplicates.

At the end, there were 6,669 arrests without a charge, and 456,300 arrests with a charge. These were joined with other .csv files which matched *RACE_CODE* → *RACE* → *RACE GROUP*. I used the *RACE* variable over *ETHNICITY* because there were less missing values.

Once these the datasets were constructed, the proportion of each *RACE GROUP* and *SEX*, as well as the distribution of *AGE*, were compared between the final dataset of arrests that did have corresponding charges versus the final dataset that did not.

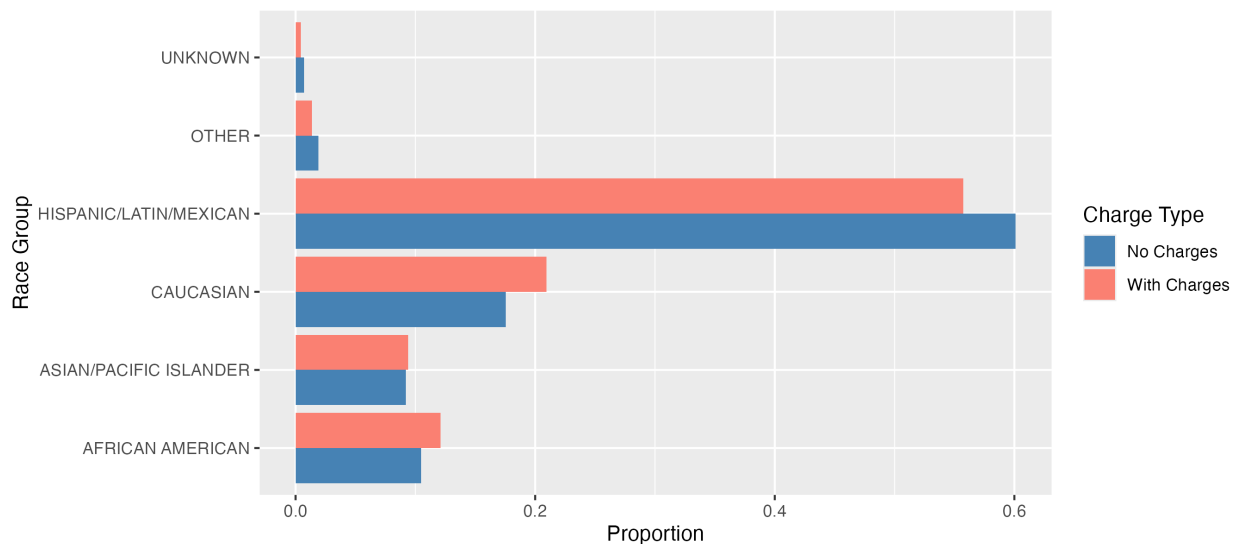
Upon investigating these proportions, a permutation test was done to conclude whether any differences in the proportions were statistically significant. The permutation test was done by sampling 6,669 rows from the dataset of arrests with corresponding charges, calculating a test

statistic, and comparing it to the actual test statistic from the real data. The test statistic was the sum of absolute proportion differences between the arrests with charges and the sampled rows by race group. This compares how likely the proportion is to change by random chance.

Finally, a look into the incidents table revealed what type of incidents are encompassing these arrests without charges.

3 Results

After aggregating the total arrests with and without charges, the proportions were displayed. The most interesting and stark difference was with *RACE GROUP*



Due to this difference a permutation test was conducted and the result was a **p-value of 0.5**. After sampling 10,000 times, about half of the time, the test statistic was at or more extreme than the true test statistic.

Although this highly suggests the phenomenon shown in the graph is due to chance, I decided to investigate the most common incidents leading to arrest without a charge. Because the racial group with the biggest difference was *HISPANIC/LATIN/MEXICAN*, I investigated this group's most popular incidents where the arrests without charges *did* have a corresponding incident.

<i>Incident</i>	<i>Frequency</i>
11377(A) HS POSN CNTRLD SUBS	277
11364 HS DRUG PARAPHERNALIA	157
23103(A) VC RECKLESS DRIVING	67
WARRANT & FUGITIVE	31
11350 HS NARCOTICS POSSESSION	27

4 Conclusion

In conclusion, only a very small fraction of arrests did not have a corresponding charge. However, there was a clear difference between the arrests that did have corresponding charges and arrests that did not in terms of racial groups.

Due to the racial makeup of San Jose itself, we cannot conclude much from either proportion alone. However in comparison, we see a weird phenomenon specifically happening to hispanic/latine/mexican people, in which they are more likely to be arrested without a charge than expected.

Most of these phenomenon (where an incident could be tracked) were drug related. Although there were a little less than a 70% of arrests from this group without a corresponding incident or charge.

An arrest without a charge may mean a mistake in the police station's handling of cases, or it could mean the arrest was faulty to begin with, so it is important to investigate these situations carefully and make sure there are no stereotypes or heuristics are employed when enforcing the law.

With more time a deeper investigation into the interactions between groups (age, sex, and race) could prove to surface more information. Also, more investigation into whether this phenomenon is happening to repeat versus first-time offenders could reveal more about these cases.

References

- [1] Justin Sherman (2021) San Jose Police Arrests: Incidents, Arrests, and Charges, October 2021. <https://www.kaggle.com/datasets/jsherman256/sjpd-incidents-arrests-charges?resource=download>